

Title: Retrieval-Augmented Medical Question Answering Assistant

Abstract:

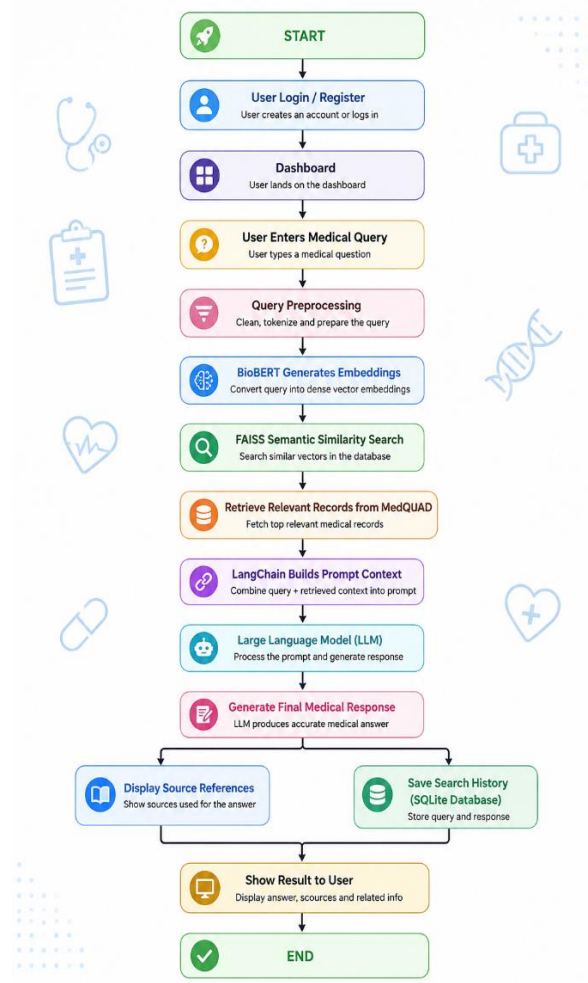
The increasing use of the internet for medical information has made it easier for people to access healthcare knowledge. However, search engines and general AI chatbots may provide inaccurate, incomplete, or unreliable medical answers, which can lead to confusion and poor healthcare decisions.

This project proposes a Retrieval-Augmented Medical Question Answering Assistant that provides accurate, reliable, and context-aware answers by retrieving information from trusted medical sources before generating responses. The system uses the MedQuAD dataset as the primary medical knowledge base and employs BioBERT to generate semantic embeddings of medical text. FAISS is used to perform semantic similarity search and retrieve the most relevant medical records based on the user's query. The retrieved information is then processed using LangChain and a Large Language Model (LLM) to generate clear and understandable medical responses through the Retrieval-Augmented Generation (RAG) approach.

The application is developed using Python, Flask, HTML, CSS, and JavaScript, providing a modern and user-friendly interface. It also includes features such as user authentication, search history, suggested questions, source references, medical disclaimers, and dark/light mode to enhance the user experience.

The proposed system aims to reduce AI hallucinations by grounding responses in trusted medical datasets and providing source-supported answers. This project demonstrates how Retrieval-Augmented Generation can improve the accuracy, reliability, and transparency of AI-powered medical question answering systems.

S.No.	Dataset	Description	Purpose in Project	URL Link
1	MedQuAD (Medical Question Answering Dataset)	Contains over 47,000 medical question-answer pairs collected from trusted NIH medical websites.	Used as the primary knowledge base for retrieving relevant medical information and generating accurate answers.	https://github.com/abachaa/MedQuAD
2	PubMed (Reference Source)	A large biomedical literature database maintained by the National Library of Medicine (NLM).	Used as a trusted reference source for validating and supporting medical information.	https://pubmed.ncbi.nlm.nih.gov/



Project Members

S. No	Name	Roll. No
1.	N. Deepika	2420030097
2.	K. Navya	2420030180
3.	K. Bhavya	2420030181